

Decision to Issue Drought Warning via Bayesian Network

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Abstract

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1 Introduction

Drought Anticipatory Action (DAA)

bnet (Bayesian network) (see Ref.[5])

Standardized Precipitation Index (SPI) ranges from -4 (least rain) to 4 (most rain).

2 Notation and Conventions

We will use $\mathbb{1}(S)$ to denote the indicator function (a.k.a. the truth function). $\mathbb{1}(S)$ equals 1 if statement S is true, and 0 if S is false. For example, $\mathbb{1}(x \geq 0)$ equals 0 if $x < 0$ and 1 otherwise.

In this article, random variables will be indicated by underlining. For example, $P(\underline{x} > x)$ will denote the probability that the random variable $\underline{x}$ is greater than x .

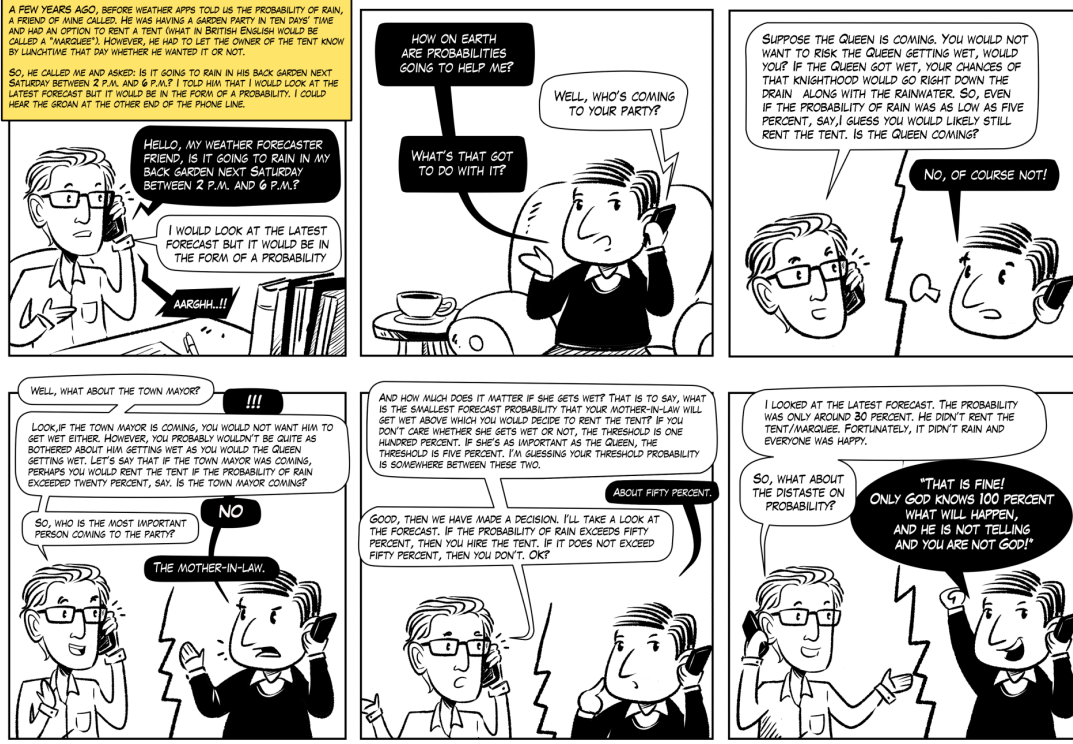
Let

N = the number of times (seasons) in which predicted and observed droughts were compared.

$N_{x|a}$ = number of events in which a number of $a \in \{0, 1\}$ droughts occurred and a number of $x \in \{0, 1\}$ droughts were predicted. For example, $N_{1|0}$ is the number of events in which no drought actually occurred but it was predicted nonetheless (False Alarm).

$N_{0|0}$ = **True Negatives (TN)**, correct rejection

$N_{1|1}$ = **True Positives (TP)**, hit



Source: Palmer, Tim. The Primacy of Doubt: From climate change to quantum physics, how the science of uncertainty can help predict and understand our chaotic world. Oxford University Press, 2022.

Figure 1: Illustration from Tim Palmer's book Ref.[4]

$N_{0|1}$ = **False Negatives** (FN), miss, type II error or underestimation

$N_{1|0}$ = **False Positives** (FP), false alarm, type I error or overestimation

$$N_{|a} = \sum_{x'} N_{x'|a}, \quad N_{x|} = \sum_{a'} N_{x|a'} \quad (1)$$

$$N = \sum_x \sum_a N_{x|a} = \sum_a N_{|a} = \sum_x N_{x|} \quad (2)$$

$$P(x|a) = R_{x|a} = \frac{N_{x|a}}{N_{|a}}, \quad P'(a|x) = R'_{a|x} = \frac{N_{x|a}}{N_{x|}} \quad (3)$$

$$\sum_x P(x|a) = \sum_a P'(a|x) = 1 \quad (4)$$

Note that both $P(x|a)$ and $P'(a|x)$ are conditional probability distributions.

Note that $N_{x|a}$ for $x, a \in \{0, 1\}$ yields 4 degrees of freedom (dofs). Instead of those 4 dofs, we will use N and 3 others dofs, namely, hit rate (HR), false alarm rate (FAR) and area under ROC curve (AUROC). HR, FAR and AUROC are defined in Appendix A

3 DAA bnet

Let

$\kappa = (\sigma, \delta)$ = the drought kind, where $\sigma \in \{\text{MAM}, \text{JJAS}\}$ ¹ is the season, and $\delta \in \{\text{Moderate}, \text{Extreme}, \text{Severe}\}$ is the drought degree.

$S_\kappa = \text{Severity}$ of drought is defined as minus SPI (higher rain, higher SPI, less severe drought). For flood (FAA), we would use $+SPI$ instead.

$P(S_\kappa)$ = likelihood, probability that a drought of severity S_κ will occur in the near future (weeks to months), based on forecasts.

μ = ensemble member index

M = number of ensemble members, 51 for *ECMWF* (European Centre for Medium-Range Weather Forecasts)

$\sigma \backslash \delta$	Moderate	Extreme	Severe
MAM	−0.55	−0.98	−0.99
JJAS	−0.41	−0.98	−0.99

Table 1: SPI threshold for Karamoja, Uganda. Cells give $-S_\kappa^*$, where $\kappa = (\sigma, \delta)$

S_κ^* = severity threshold for drought (See Table 3 for examples of S_κ^* .)

$$P(\underline{S}_\kappa \geq S_\kappa^*) = \frac{1}{M} \sum_{\mu=1}^M \mathbb{1}(S^{\kappa, \mu} \geq S_\kappa^*) \quad (5)$$

Let $X_\kappa^i \in \{0, 1\}$ for $i = 1, 2, 3, \dots, nx$ represent a flag for local problems such as high road density, high population density, historical catastrophic events, etc. It equals 1 if a particular danger is present, and 0 if not.

Let

$\epsilon \in \mathbb{R}$ be a normally distributed random variable with mean zero and some a priori given deviation.

RC_κ^i = replacement cost for flag X_κ^i

IC_κ^i = insurance (i.e., preventive) cost for flag X_κ^i

Note that insurance always costs more than replacement, so

$$0 \leq IC_\kappa^i \leq RC_\kappa^i \quad (6)$$

$P_{danger, \kappa}^i = \frac{IC_\kappa^i}{RC_\kappa^i}$ probability of danger for flag X_κ^i .

Note that less danger means less insurance cost.

P_κ^* = probability threshold for drought

$$P_\kappa^* = f_\kappa(X_\kappa^1, X_\kappa^2, \dots, X_\kappa^{nx}) + \epsilon \quad (7)$$

where we model the function $f_\kappa()$ by

$$f_\kappa(X_\kappa^1, X_\kappa^2, \dots, X_\kappa^{nx}) = 1 - \max_i \{P_{danger, \kappa}^i X_\kappa^i\} \quad (8)$$

¹Letters here stand for the first letter of the months of the year.

W_κ = drought warning, either 0 or 1

P_κ^* = threshold probability that triggers a drought warning

$$W_\kappa = \mathbb{1} [P(\underline{S}_\kappa \geq S_\kappa^*) > P_\kappa^*] \quad (9)$$

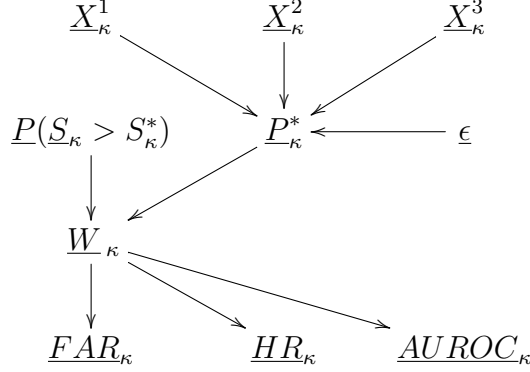


Figure 2: DAA bnet. We use one of these bnets for each κ value and for each grid cell of 0.25 degrees in latitude by 0.25 degrees in longitude. 1 degree in latitude or longitude is approximately 100 km.

The TPMs² printed in blue, for the bnet Fig.2, are as follows.³

$$P_\kappa^* = f_\kappa(X_\kappa^1, X_\kappa^2, \dots, X_\kappa^{nx}) + \epsilon \quad (10)$$

$$W_\kappa = \mathbb{1} [P(\underline{S}_\kappa > S_\kappa^*) > P_\kappa^*] \quad (11)$$

$$P(Y_\kappa^i | W_\kappa) = \text{to be learned empirically from dataset} \quad (12)$$

where $(Y_\kappa^1, Y_\kappa^2, Y_\kappa^3) = (FAR_\kappa, HR_\kappa, AUROC_\kappa)$.

Algorithm for finding best triggers T_κ^* :

Enter bnet Fig.2 into bnet software PyAgrum (Ref.[1]). Enter into software available evidence for nodes $P(\underline{S}_\kappa > S_\kappa^*)$, (X^1, X^2, X^3) and $(Y_\kappa^1, Y_\kappa^2, Y_\kappa^3)$. Allow software to infer the distribution of P_κ^* . Choose for P_κ^* the state with maximum probability.

²TPM=Transition Probability Matrix. This is the same thing as CPT=Conditional Probability Table

³A deterministic node satisfying the structural equation $y = f(x_1, x_2)$, has the TPM: $P(y|x_1, x_2) = \delta(y, f(x_1, x_2))$, where $\delta(a, b) = \mathbb{1}(a = b)$ is the Kronecker delta function.

4 Relation to Climada

This project used the Climada (see Ref.[8]) software library. Table 2 relates their terminology to ours.

In Climada, a hazard model yields the intensity for a particular event.

Climada terminology	Our terminology
Intensity of event	S =severity of event
Vulnerability= $\frac{Impact}{Exposure}$	$P_{danger} = \frac{IC}{RC}$ =danger probability
Impact	IC = insurance cost (\$)
Exposure	RC =replacement cost (\$)

Table 2: Relationship between Climada terminology and ours

Climada takes as input the intensity of an event, and calculates from this the event's vulnerability, impact and exposure. In this project, we use Climada for each flag X^i .

5 Risk

$$Risk_{\kappa} = E[IC_{\kappa}] = \frac{1}{M} \sum_{\mu=1}^M IC^{\kappa, \mu} \quad (13)$$

$$= \sum_{IC_{\kappa}} P(IC_{\kappa}) IC_{\kappa} \quad (14)$$

A **risk matrix** is a 4×4 matrix with X axis equal to insurance cost (IC) (a.k.a. impact) and Y axis equal to likelihood P , with IC binned into four values and P binned into 4 values. The matrix elements give the value of some arbitrary function $f(IC, P)$ defined as risk. Usually, $f(IC, P) = IC * P$ but not necessarily. I find risk matrices arbitrary and not very useful. The Wikipedia article on risk matrices (Ref.[7]) agrees.

6 Damage Functions

Bring the following disaster prediction models together via Damage Function

- (a) Ref.[4] book by Tim Palmer
- (b) Ref.[8] Climada Software
- (c) Ref.[2] by Huizinga et al. Global flood depth damage functions
- (d) Ref.[3] by Liu et al. A method of deriving damage fractions functions.

α = index that ranges over different hazards

$f_\alpha(x_\alpha)$ = **Damage Fraction** (term used in Ref.[2]), **Impact Function** (term used in Ref.[8]) for Hazard α . $DF_\alpha(x_\alpha)$ is a function of the parameter x_α , and increases monotonically from 0 when $x_\alpha = 0$ to 1 when x_α is maximum, $x_\alpha = x_{\alpha,max}$

$D_{\alpha,max}$ = **Maximum Damage** (term used in Ref.[2]), **Exposure** (term used in Ref.[8]) for Hazard α . In dollars.

Henceforth, anything called a Damage will be in units of dollars (or some other currency), except damage fractions, which will be unitless, and range from zero to one.

$D_\alpha(x_\alpha)$ = **Damage** (term used in Ref.[2]), **Impact** (term used in Ref.[8])

$$D_\alpha(x_\alpha) = D_{\alpha,max} f_\alpha(x_\alpha) \quad (15)$$

$\bar{D}_\alpha$ = **Average Damage**

$$\bar{D}_\alpha = \frac{1}{x_{\alpha,max}} \int_0^{x_{\alpha,max}} dx f_\alpha(x_\alpha) \quad (16)$$

D_{max} = **maximum total damage** that is tolerable. Any time total Damage is higher than this, always issue drought warning.

We assume x_α is binned and $x_\alpha \geq 0$. However, we will use $x_\alpha = \text{None}$ to indicate

$$D_\alpha(x_\alpha = \text{None}) = \bar{D}_\alpha \quad (17)$$

When $x_\alpha \neq \text{None}$, it should be the value of x_α for some historical event.

ϵ_α = **error** in D_α , normally distributed with mean zero and standard deviation σ_α .

$$D'_\alpha(x_\alpha) = D_\alpha(x_\alpha) + \epsilon_\alpha \quad (18)$$

f_D = **total Damage fraction**

$$f_D = \begin{cases} 1 & \text{if } \sum_\alpha D'_\alpha(x_\alpha) > D_{max} \\ \frac{1}{D_{max}} \sum_\alpha D'_\alpha(x_\alpha) & \text{otherwise} \end{cases} \quad (19)$$

Let

$\kappa = (\sigma, \delta)$ = the drought kind, where $\sigma \in \{\text{MAM, JJAS}\}^4$ is the season, and $\delta \in \{\text{Moderate, Extreme, Severe}\}$ is the drought degree.

S_κ = **Severity** of drought is defined as minus SPI (higher rain, higher SPI, less severe drought). For flood (FAA), we would use $+SPI$ instead. Severity is called **Intensity** in Ref.[8]

$P(S_\kappa)$ = likelihood, probability that a drought of severity S_κ will occur in the near future (weeks to months), based on forecasts.

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$\sigma \backslash \delta$	Moderate	Extreme	Severe
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Table 3: SPI threshold for Karamoja, Uganda. Cells give $-S_\kappa^*$, where $\kappa = (\sigma, \delta)$

M = number of ensemble members, 51 for *ECMWF* (European Centre for Medium-Range Weather Forecasts)

S_κ^* = severity threshold for drought (See Table 3 for examples of S_κ^* .)

$$P(\underline{S}_\kappa \geq S_\kappa^*) = \frac{1}{M} \sum_{\mu=1}^M \mathbb{1}(S^{\kappa, \mu} \geq S_\kappa^*) \quad (20)$$

W_κ = **drought warning**, either 0 or 1

$f_{D, \kappa}$ = the total damage fraction f_D defined previously, for drought kind κ .

We will use this as a threshold for triggering a drought warning

$$W_\kappa = \mathbb{1} [P(\underline{S}_\kappa \geq S_\kappa^*) > 1 - f_{D, \kappa}] \quad (21)$$

Bayesian Networks

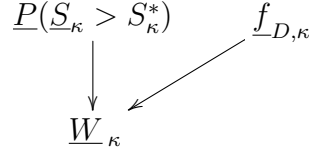


Figure 3: bnet for calculating warning W_κ

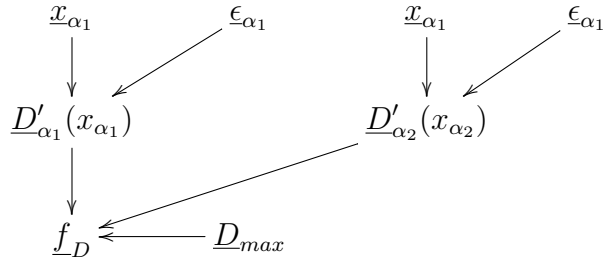


Figure 4: bnet for calculating total damage fraction f_D for two hazards α_1 and α_2 .

⁴Letters here stand for the first letter of the months of the year.

A Appendix: Functions of $N_{x|a}$

Terminology Table Adapted from Wikipedia Ref.[6]

positive events (P): $P = N_{|1} = \sum_x N_{x|1}$, the number of real positive events in the data

negative events (N): $N = N_{|0} = \sum_x N_{x|0}$, the number of real negative events in the data

sensitivity, recall, true positive rate (TPR), or hit rate (**HR**):

$$TPR = R_{1|1} = \frac{N_{1|1}}{N_{|1}} = \frac{N_{1|1}}{N_{1|1} + N_{0|1}} = 1 - R_{0|1} \quad (22)$$

specificity, selectivity or true negative rate (TNR):

$$TNR = R_{0|0} = \frac{N_{0|0}}{N_{|0}} = \frac{N_{0|0}}{N_{0|0} + N_{1|0}} = 1 - R_{1|0} \quad (23)$$

precision or positive predictive value (PPV):

$$PPV = \frac{N_{1|1}}{N_{1|1} + N_{1|0}} = 1 - FDR \quad (24)$$

negative predictive value (NPV):

$$NPV = \frac{N_{0|0}}{N_{0|0} + N_{0|1}} = 1 - FOR \quad (25)$$

miss rate or false negative rate (FNR):

$$FNR = R_{0|1} = \frac{N_{0|1}}{N_{|1}} = \frac{N_{0|1}}{N_{0|1} + N_{1|1}} = 1 - R_{1|1} \quad (26)$$

fall-out, false positive rate (FPR), or false alarm rate (**FAR**):

$$FPR = R_{1|0} = \frac{N_{1|0}}{N_{|0}} = \frac{N_{1|0}}{N_{1|0} + N_{0|0}} = 1 - R_{0|0} \quad (27)$$

false discovery rate (FDR):

$$FDR = \frac{N_{1|0}}{N_{1|0} + N_{1|1}} = 1 - PPV \quad (28)$$

false omission rate (FOR):

$$FOR = \frac{N_{0|1}}{N_{0|1} + N_{0|0}} = 1 - NPV \quad (29)$$

accuracy (ACC):

$$ACC = \frac{N_{1|1} + N_{0|0}}{N_{|1} + N_{|0}} = \frac{N_{1|1} + N_{0|0}}{N_{1|1} + N_{0|0} + N_{1|0} + N_{0|1}} \quad (30)$$

balanced accuracy (BA):

$$BA = \frac{R_{1|1} + R_{0|0}}{2} \quad (31)$$

F1 score is the harmonic mean of precision and sensitivity:

$$F_1 = 2 \times \frac{PPV \times R_{1|1}}{PPV + R_{1|1}} = \frac{2N_{1|1}}{2N_{1|1} + N_{1|0} + N_{0|1}} \quad (32)$$

The area under the ROC (Receiver Operating Characteristic) curve, known as AUC or **AUROC**, is defined by Fig.5. One set of 4 values $N_{x|a}$ where $a, x \in \{0, 1\}$, only gives one point of the ROC curve. To get a full ROC curve, we need to vary a parameter. In the case of DAA, we vary geographical location, for example.

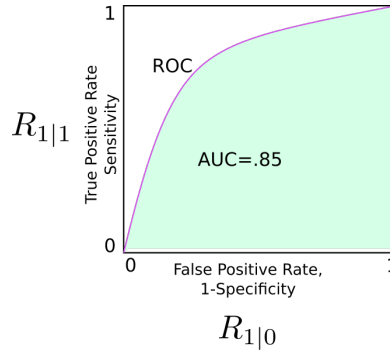


Figure 5: Example of ROC. Green shaded area is the AUC of the ROC.

References

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